POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Spring

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write the mechanism of buffer action of a solution containing a mixture of benzoic acid and sodium benzoate. Calculate the PH of 500ml of a buffer solution containing 0.2M ammonium sulphate and 0.3M ammonia which is 2.1% ionized in dilute solution.

   b) What do you mean by standard electrode potential? How single electrode is potential originated? Calculate the emf of the following cell at 25^0 C $Zn/Zn^{++}(0.01M)//\ Cu^{++}(0.1M)/Cu$.

   Given,

   $E^0\ Zn/Zn^{++} = 0.76V$

   $E^0\ cu/cu^{++} = -0.34V$

   $R = 8.314\ JK^{-1}\ mol^{-1}$, $F = 96500C$

2. a) Give reasons

   i. Transition elements form significant number of complexes.

   ii. $TiCl_3$ compounds are coloured but those of $TiCl_4$ are colorless.

   iii. Zn is not considered as true transition elements

   iv. Transition elements show variable valency.

   b) Define Ionization Energy. Explain the factors affecting electro negativity? Why electron affinity of chlorine is higher than fluorine?

3. a) How do enantiomers differ from diasteriomers? Differentiate between carbocations and carbonanions within their stability.

   b) What are elimination reactions? Write the mechanism of E_1 and E_2 reactions giving suitable examples.

4. a) Explain addition polymerization. How it differs from condensation polymerization? 7

b) Write short notes on :

i. Vulcanization of rubber. 8

ii. Nylon 6,6

OR

Explain de-Broglie's Principle of dual nature of electron. Calculate the radius of the orbit of electron of Hydrogen atom which is in the 3rd energy level.(mass of electron=9.1091×10^{-31}kg, Plank's constant, h=6.62×10^{-34}Js, permittivity, to= 8.85×10^{-12} $kg^{-1}m^{-3}A^{2}$,charge on electron, e=1.6×10^{-19}C)

5. a) What do you mean by Paper chromatography? Explain briefly, how it can be used for analyzing solutes present in the given sample 7

OR

Explain the following:

i. The compound having metallic bonds are good conductor of heat and electricity. 7

ii. Covalent bonds are directional in nature

iii. H_2O exist as liquid but H_{25} as gas at room temperature

iv. Contraction of water takes place up to 4^0c

b) What are lubricants? Give its functions. Give the preparation method and used of TNT. 8

OR

Differentiate between Molecular Orbital Theory with Valence Bond Theory. Write the characteristics of S- block elements. 8

a) What are the causes of air pollution? Write its impact on human health. Also mention its controlling measures. 7

b) Define hardness of water. How it is estimated in the laboratory? 8

Write short notes on: **(Any Two)** 2×5

a) Corrosion

b) Ozone layer depletion

c) Green house effect and global warming.